Centre for Artificial Intelligence

B. Tech. III Semester *(Artificial Intelligence (AI))*

(for batch admitted in academic session 2025-26)

S. No.	Course Code	Category Code	Course Name	Maximum Marks Allotted					Total Marks	Contact Hours per week			Total Credits	Mode of Learning	Mode of Major Evaluation	Duration of Major Evaluation
				Theory Block			Practical Block									
				Continuous Evaluation		Major Evaluation	Continuous Evaluation	Major Evaluation		L	T	P				
				Minor Evaluation	Quiz/ Assignment###		Lab Work & Sessional									
1.	31252101	BSC	Probability and Statistics	30	40	30	-	-	100	3	1	-	4	Face to Face	MCQ	1.5 Hrs
2.	31252102	DC	Design & Analysis of Algorithms	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
3.	31252103	DC	Database Management System	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
4.	31252104	DC	Computer Networks	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
5.	31252105	DC	Operating Systems	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
6.	31252106	DLC	Problem Solving through Python Programming	-	-	-	70	30	100	-	-	2	1	Experimental	AO	-
7.	31252107	DLC	Design & Analysis of Algorithms Lab	-	-	-	70	30	100	-	-	2	1	Experimental	AO	-
8.	31252108	SP	Semester Proficiency$	-	-	-	50	-	50	-	-	2	1	Face to Face	SO	-
9.	31252109	PBL	Macro Project-I#	-	-	-	70	30	100	-	-	2	1	Experiential	SO	-
10.	31252110	SLP	Self-learning/Presentation$$$ (SWAYAM/NPTEL/MOOC)	-	-	-	40	-	40	-	-	2	1	Mentoring	SO	-
11.	NECXXXXX	NEC	Novel Engaging Course (Activity Based Learning)	-	-	-	50	-	50	-	1	-	1	Interactive	SO	-
Total				**150**	**200**	**150**	**350**	**90**	**940**	**11**	**06**	**10**	**22**	-	-	-
12.	31252111	MAC	Cyber Security	-	-	-	100	-	100	-	2	-	GRADE	Interactive	SO	-
13.	31252112	MWS	Mandatory Workshop as per Annexure at Department Level										GRADE	Interactive	MCQ	-
Skill Internship Program (Institute Level) (Qualifier): Minimum 30 hours duration: To be Credited in IV Semester																

List of Courses for Self-learning/Presentation (SWAYAM/NPTEL/MOOC)			
1	Demystifying Networking (4 weeks)	3	Technical Communication for Engineers (4 weeks)
2	Mobile Virtual Reality and Artificial Intelligence (4 weeks)	4	Gender Justice and Workplace Security (4 weeks)

$Semester Proficiency– includes the weightage towards ability/ skill/ competency /knowledge level /expertise attained etc. in the semester courses

MCQ: Multiple Choice Question **AO:** Assignment + Oral **PP:** Pen Paper **SO:** Submission + Oral **OB:** Open Book

Macro Project-I will be presented and evaluated through an interdisciplinary project evaluation committee. It will also include **Database Management System Lab.**

$$$ Compulsory registration for one online course using SWAYAM/NPTEL/ MOOC, evaluation through attendance and presentation.

Quiz/Assignment of 40 marks will be evaluated out of two 20 marks components as per the guidelines of the academic section.

HSMC	BSC	ESC	DC	DE	SPC	OC	DLC	NEC	SP	SIP	SLP	PDC	PBL	MAC	MWS
0	1	0	4	0	0	0	2	1	1	0	1	0	1	1	1

Mode of Learning					Mode of Examination				Total Credits
Face to Face	Interactive	Mentoring	Experiential	Experimental	PP	AO	MCQ	SO	
17	1	1	2	1	4.8	2	11.2	4	**22**
77.27	**4.55**	**4.55**	**9.08**	**4.55**	**21.82**	**9.08**	**50.91**	**18.19**	**Credit %**

Recommended in the Board of Studies Meeting of Centre for Artificial Intelligence held on 1st June, 2026

Centre for Artificial Intelligence

B. Tech. III Semester *(Information Technology (Artificial Intelligence and Robotics))*

(for batch admitted in academic session 2025-26)

S. No.	Course Code	Category Code	Course Name	Maximum Marks Allotted					Total Marks	Contact Hours per week			Total Credits	Mode of Learning	Mode of Major Evaluation	Duration of Major Evaluation
				Theory Block			Practical Block									
				Continuous Evaluation		Major Evaluation	Continuous Evaluation	Major Evaluation		L	T	P				
				Minor Evaluation	Quiz/ Assignment###		Lab Work & Sessional									
1.	24252101	BSC	Probability and Statistics	30	40	30	-	-	100	3	1	-	4	Face to Face	MCQ	1.5 Hrs
2.	24252102	DC	Design & Analysis of Algorithms	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
3.	24252103	DC	Control System	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
4.	24252104	DC	Computer Networks	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
5.	24252105	DC	Operating Systems	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
6.	24252106	DLC	Problem Solving through Python Programming	-	-	-	70	30	100	-	-	2	1	Experimental	AO	-
7.	24252107	DLC	Design & Analysis of Algorithms Lab	-	-	-	70	30	100	-	-	2	1	Experimental	AO	-
8.	24252108	SP	Semester Proficiency$	-	-	-	50	-	50	-	-	2	1	Face to Face	SO	-
9.	24252109	PBL	Macro Project-I#	-	-	-	70	30	100	-	-	2	1	Experiential	SO	-
10.	24252110	SLP	Self-learning/Presentation$$$ (SWAYAM/NPTEL/MOOC)	-	-	-	40	-	40	-	-	2	1	Mentoring	SO	-
11.	NECXXXXX	NEC	Novel Engaging Course (Activity Based Learning)	-	-	-	50	-	50	-	1	-	1	Interactive	SO	-
Total				**150**	**200**	**150**	**350**	**90**	**940**	**11**	**06**	**10**	**22**	-	-	-
12.	24252111	MAC	Cyber Security	-	-	-	100	-	100	-	2	-	GRADE	Interactive	SO	-
13.	24252112	MWS	Mandatory Workshop as per Annexure at Department Level										GRADE	Interactive	MCQ	-
Skill Internship Program (Institute Level) (Qualifier): Minimum 30 hours duration: To be Credited in IV Semester																

List of Courses for Self-learning/Presentation (SWAYAM/NPTEL/MOOC)			
1	Foundations of Cognitive Robotics (4 weeks)	3	Technical Communication for Engineers (4 weeks)
2	Mobile Virtual Reality and Artificial Intelligence (4 weeks)	4	Gender Justice and Workplace Security (4 weeks)

$Semester Proficiency– includes the weightage towards ability/ skill/ competency /knowledge level /expertise attained etc. in the semester courses

MCQ: Multiple Choice Question **AO:** Assignment + Oral **PP:** Pen Paper **SO:** Submission + Oral **OB:** Open Book

Macro Project-I will be presented and evaluated through an interdisciplinary project evaluation committee. It will also include the **Design Thinking Lab.**

$$$ Compulsory registration for one online course using SWAYAM/NPTEL/ MOOC, evaluation through attendance and presentation.

Quiz/Assignment of 40 marks will be evaluated out of two 20 marks components as per the guidelines of the academic section.

HSMC	BSC	ESC	DC	DE	SPC	OC	DLC	NEC	SP	SIP	SLP	PDC	PBL	MAC	MWS
0	1	0	4	0	0	0	2	1	1	0	1	0	1	1	1

Mode of Learning					Mode of Examination				Total Credits
Face to Face	Interactive	Mentoring	Experiential	Experimental	PP	AO	MCQ	SO	
17	1	1	2	1	4.8	2	11.2	4	**22**
77.27	**4.55**	**4.55**	**9.08**	**4.55**	**21.82**	**9.08**	**50.91**	**18.19**	**Credit %**

Recommended in the Board of Studies Meeting of Centre for Artificial Intelligence held on 1st June, 2026

Centre for Artificial Intelligence

B. Tech. III Semester *(Artificial Intelligence (AI) and Data Science)*

(for batch admitted in academic session 2025-26)

S. No.	Course Code	Category Code	Course Name	Maximum Marks Allotted					Total Marks	Contact Hours per week			Total Credits	Mode of Learning	Mode of Major Evaluation	Duration of Major Evaluation
				Theory Block			Practical Block									
				Continuous Evaluation		Major Evaluation	Continuous Evaluation	Major Evaluation		L	T	P				
				Minor Evaluation	Quiz/ Assignment###		Lab Work & Sessional									
1.	27252101	BSC	Probability and Statistics	30	40	30	-	-	100	3	1	-	4	Face to Face	MCQ	1.5 Hrs
2.	27252102	DC	Design & Analysis of Algorithms	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
3.	27252103	DC	Database Management System	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
4.	27252104	DC	Computer Networks	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
5.	27252105	DC	Operating Systems	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
6.	27252106	DLC	Problem Solving through Python Programming	-	-	-	70	30	100	-	-	2	1	Experimental	AO	-
7.	27252107	DLC	Design & Analysis of Algorithms Lab	-	-	-	70	30	100	-	-	2	1	Experimental	AO	-
8.	27252108	SP	Semester Proficiency$	-	-	-	50	-	50	-	-	2	1	Face to Face	SO	-
9.	27252109	PBL	Macro Project-I#	-	-	-	70	30	100	-	-	2	1	Experiential	SO	-
10.	27252110	SLP	Self-learning/Presentation$$$ (SWAYAM/NPTEL/MOOC)	-	-	-	40	-	40	-	-	2	1	Mentoring	SO	-
11.	NECXXXXX	NEC	Novel Engaging Course (Activity Based Learning)	-	-	-	50	-	50	-	1	-	1	Interactive	SO	-
Total				**150**	**200**	**150**	**350**	**90**	**940**	**11**	**06**	**10**	**22**	-	-	-
12.	27252111	MAC	Cyber Security	-	-	-	100	-	100	-	2	-	GRADE	Interactive	SO	-
13.	27252112	MWS	Mandatory Workshop as per Annexure at Department Level										GRADE	Interactive	MCQ	-
Skill Internship Program (Institute Level) (Qualifier): Minimum 30 hours duration: To be Credited in IV Semester																

List of Courses for Self-learning/Presentation (SWAYAM/NPTEL/MOOC)			
1	Demystifying Networking (4 weeks)	3	Technical Communication for Engineers (4 weeks)
2	Mobile Virtual Reality and Artificial Intelligence (4 weeks)	4	Gender Justice and Workplace Security (4 weeks)

$Semester Proficiency– includes the weightage towards ability/ skill/ competency /knowledge level /expertise attained etc. in the semester courses

MCQ: Multiple Choice Question **AO:** Assignment + Oral **PP:** Pen Paper **SO:** Submission + Oral **OB:** Open Book

Macro Project-I will be presented and evaluated through an interdisciplinary project evaluation committee. It will also include **Database Management System Lab**

$$$ Compulsory registration for one online course using SWAYAM/NPTEL/ MOOC, evaluation through attendance and presentation.

Quiz/Assignment of 40 marks will be evaluated out of two 20 marks components as per the guidelines of the academic section.

HSMC	BSC	ESC	DC	DE	SPC	OC	DLC	NEC	SP	SIP	SLP	PDC	PBL	MAC	MWS
0	1	0	4	0	0	0	2	1	1	0	1	0	1	1	1

Mode of Learning					Mode of Examination				Total Credits
Face to Face	Interactive	Mentoring	Experiential	Experimental	PP	AO	MCQ	SO	
17	1	1	2	1	4.8	2	11.2	4	**22**
77.27	**4.55**	**4.55**	**9.08**	**4.55**	**21.82**	**9.08**	**50.91**	**18.19**	**Credit %**

Recommended in the Board of Studies Meeting of Centre for Artificial Intelligence held on 1st June, 2026

Centre for Artificial Intelligence

B. Tech. III Semester *(Artificial Intelligence (AI) and Machine Learning)*

(for batch admitted in academic session 2025-26)

S. No.	Course Code	Category Code	Course Name	Maximum Marks Allotted					Total Marks	Contact Hours per week			Total Credits	Mode of Learning	Mode of Major Evaluation	Duration of Major Evaluation
				Theory Block			Practical Block									
				Continuous Evaluation		Major Evaluation	Continuous Evaluation	Major Evaluation								
				Minor Evaluation	Quiz/ Assignment###		Lab Work & Sessional			L	T	P				
1.	28252101	BSC	Probability and Statistics	30	40	30	-	-	100	3	1	-	4	Face to Face	MCQ	1.5 Hrs
2.	28252102	DC	Design & Analysis of Algorithms	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
3.	28252103	DC	Database Management System	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
4.	28252104	DC	Computer Networks	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
5.	28252105	DC	Operating Systems	30	40	30	-	-	100	2	1	-	3	Face to Face	MCQ	1.5 Hrs
6.	28252106	DLC	Problem Solving through Python Programming	-	-	-	70	30	100	-	-	2	1	Experimental	AO	-
7.	28252107	DLC	Design & Analysis of Algorithms Lab	-	-	-	70	30	100	-	-	2	1	Experimental	AO	-
8.	28252108	SP	Semester Proficiency$	-	-	-	50	-	50	-	-	2	1	Face to Face	SO	-
9.	28252109	PBL	Macro Project-I#	-	-	-	70	30	100	-	-	2	1	Experiential	SO	-
10.	28252110	SLP	Self-learning/Presentation$$$ (SWAYAM/NPTEL/MOOC)	-	-	-	40	-	40	-	-	2	1	Mentoring	SO	-
11.	NECXXXXX	NEC	Novel Engaging Course (Activity Based Learning)	-	-	-	50	-	50	-	1	-	1	Interactive	SO	-
Total				**150**	**200**	**150**	**350**	**90**	**940**	**11**	**06**	**10**	**22**	-	-	-
12.	28252111	MAC	Cyber Security	-	-	-	100	-	100	-	2	-	GRADE	Interactive	SO	-
13.	28252112	MWS	Mandatory Workshop as per Annexure at Department Level										GRADE	Interactive	MCQ	-
Skill Internship Program (Institute Level) (Qualifier): Minimum 30 hours duration: To be Credited in IV Semester																

List of Courses for Self-learning/Presentation (SWAYAM/NPTEL/MOOC)			
1	Demystifying Networking (4 weeks)	3	Technical Communication for Engineers (4 weeks)
2	Mobile Virtual Reality and Artificial Intelligence (4 weeks)	4	Gender Justice and Workplace Security (4 weeks)

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HSMC	BSC	ESC	DC	DE	SPC	OC	DLC	NEC	SP	SIP	SLP	PDC	PBL	MAC	MWS
0	1	0	4	0	0	0	2	1	1	0	1	0	1	1	1

Mode of Learning					Mode of Examination				Total Credits
Face to Face	Interactive	Mentoring	Experiential	Experimental	PP	AO	MCQ	SO	
17	1	1	2	1	4.8	2	11.2	4	**22**
77.27	**4.55**	**4.55**	**9.08**	**4.55**	**21.82**	**9.08**	**50.91**	**18.19**	**Credit %**

Recommended in the Board of Studies Meeting of Centre for Artificial Intelligence held on 1st June, 2026